Form D

Core Focus on Proportions & Probability

Block 1 Review ~ Ratios and Rates

ANSWER KEY

Name _____ Period _____ Date _____

Part I – Selected Response

1. Complete the conversion:
4 yards = _____ inches

A. 12

B. 48

C. 144

D. 432

C (two steps: 4 yd × 3 = 12 ft, then 12 ft × 12 = 144 in)

2. Which numbers are equivalent to 2.75? Circle all that apply.

A. $2\frac{3}{4}$

B. $\frac{11}{4}$

C. $2\frac{7}{10}$

D. $\frac{4}{11}$

E. $2\frac{75}{100}$

F. $\frac{275}{10}$

A, B, E ($0.75 = \frac{75}{100} = \frac{3}{4}$; $2 \times 4 + 3 = 11$, so $\frac{11}{4}$. C is 2.7, D is upside down, F is 27.5)

3. Which list shows the ratios in order from least to greatest?

A. $\frac{4}{5}$, 3 : 4, 7 to 10

B. 3 : 4, 7 to 10, $\frac{4}{5}$

C. 7 to 10, 3 : 4, $\frac{4}{5}$

D. 7 to 10, $\frac{4}{5}$, 3 : 4

C (as decimals: 7 to 10 = 0.7, 3 : 4 = 0.75, $\frac{4}{5}$ = 0.8)

4. A hockey team has a win to loss ratio of 3 : 4. Which could be the team's record?

A. 6 wins and 14 games played

B. 8 wins and 6 losses

C. 9 wins and 12 games played

D. 12 wins and 15 losses

A (14 – 6 = 8 losses; 6 : 8 = 3 : 4. B is 4 : 3, C is 9 : 3 = 3 : 1, D is 4 : 5)

5. A store sells packs of granola bars. Which pack is the best buy?

A. 6 bars for $4.80

B. 10 bars for $7.50

C. 15 bars for $11.25

D. 12 bars for $8.40

D (price per bar: A $0.80, B $0.75, C $0.75, D $8.40 ÷ 12 = $0.70, the lowest)

6. Which rate is **not** equal to 1?

A. $\frac{12 \text{ inches}}{1 \text{ foot}}$

B. $\frac{1 \text{ gallon}}{4 \text{ quarts}}$

C. $\frac{1 \text{ pound}}{12 \text{ ounces}}$

D. $\frac{1{,}000 \text{ meters}}{1 \text{ kilometer}}$

C (1 pound = 16 ounces, not 12, so the top and bottom are not the same amount)

7. For numbers 7a – 7d, circle YES or NO to tell whether the rate given is equivalent to a conveyor belt moving 18 feet per minute.

7a. 6 yards per minute	YES	NO
7b. 216 inches per minute	YES	NO
7c. 1,080 feet per second	YES	NO
7d. 1,080 feet per hour	YES	NO

7a YES (18 ÷ 3 = 6) · 7b YES (18 × 12 = 216) · 7c NO (a second is shorter than a minute, so the number should be smaller: 18 ÷ 60 = 0.3 ft per s) · 7d YES (18 × 60 = 1,080)

8. Simplify the complex fraction $\frac{6}{\frac{3}{4}}$.

A. $4\frac{1}{2}$

B. 8

C. $\frac{1}{8}$

D. $\frac{3}{24}$

B ($6 \div \frac{3}{4} = \frac{6}{1} \times \frac{4}{3} = \frac{24}{3} = 8$)

9. Tess rode her bike $2\frac{1}{2}$ miles in 15 minutes. Assume she rode the entire time at the same rate. What was her speed in miles per hour?

A. 10 miles per hour

B. 6 miles per hour

C. $\frac{1}{6}$ mile per hour

D. $37\frac{1}{2}$ miles per hour

A (15 minutes = $\frac{1}{4}$ hour; $\frac{5}{2} \div \frac{1}{4} = \frac{5}{2} \times \frac{4}{1} = 10$ mph)

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Part II – Constructed Response

Complete each conversion.

1. 5 gallons = _____ quarts

20 quarts (5 × 4)

2. 3 kilometers = _____ centimeters

300,000 centimeters (3 × 1,000 = 3,000 m, then 3,000 × 100 = 300,000 cm)

3. 36 fluid ounces = _____ cups

4.5 cups (36 ÷ 8)

4. 420 seconds = _____ minutes

7 minutes (420 ÷ 60)

Convert each mixed number to a decimal.

5. $2\frac{5}{8}$

2.625 (keep the 2; 5 ÷ 8 = 0.625)

6. $3\frac{2}{9}$

$3.\overline{2}$ (keep the 3; 2 ÷ 9 = 0.222…)

Convert each decimal to a fraction or mixed number in simplest form.

7. 0.85

$\frac{17}{20}$ ($\frac{85}{100} \div 5 = \frac{17}{20}$)

8. 3.125

$3\frac{1}{8}$ ($3\frac{125}{1000} \div 125 = 3\frac{1}{8}$)

9. Simplify the ratio 30 : 42. Write the simplified ratio in all three forms.

$\frac{5}{7}$, 5 : 7, 5 to 7 (divide both by 6)

10. Fourteen of 21 tomato plants in a community garden have ripe tomatoes. Write three different simplified number ratios using this information. Explain the meaning of each ratio using words.

Ratio (numbers):	Ratio (numbers):	Ratio (numbers):
Explanation: 2 : 3 — plants with ripe tomatoes to all plants (14 : 21)	Explanation: 1 : 3 — plants without ripe tomatoes to all plants (7 : 21)	Explanation: 2 : 1 — plants with ripe tomatoes to plants without (14 : 7). The reversed ratios 3 : 2, 3 : 1 and 1 : 2 are also correct if explained.

Find the unit rate.

11. Liam read 30 pages in 45 minutes. Write his unit rate in pages per minute. Then write it in minutes per page.

$\frac{2}{3}$ page per minute ($\frac{30}{45} = \frac{2}{3}$); $1\frac{1}{2}$ minutes per page ($\frac{45}{30} = \frac{3}{2}$)

12. $\frac{3\frac{1}{3} \text{ miles}}{\frac{5}{6} \text{ hour}}$

4 miles per hour ($\frac{10}{3} \times \frac{6}{5} = \frac{60}{15} = 4$)

13. A 24-ounce jar of peanut butter costs $4.32. A 40-ounce jar costs $6.80. Which jar is the better buy? Show all work necessary to justify your answer.

24-oz jar: $4.32 ÷ 24 = $0.18 per ounce. 40-oz jar: $6.80 ÷ 40 = $0.17 per ounce. The 40-ounce jar is the better buy.

14. A jogger runs at a rate of 6 miles per hour. Write this rate as feet per second. Show all work necessary to justify your answer.

8.8 feet per second ($\frac{6 \text{ mi}}{1 \text{ hr}} \times \frac{5{,}280 \text{ ft}}{1 \text{ mi}} = \frac{31{,}680 \text{ ft}}{1 \text{ hr}}$; 1 hr = 3,600 s, so 31,680 ÷ 3,600 = 8.8 ft per s)

15. Ava swam 1.2 kilometers in 30 minutes. Ben swims at a rate of 45 meters per minute. Who swims faster? Show all work necessary to justify your answer.

Ava: 1.2 km × 1,000 = 1,200 m; 1,200 m ÷ 30 min = 40 meters per minute. Ben: 45 meters per minute. Ben swims faster.

16. Owen walked $2\frac{1}{4}$ miles in $\frac{3}{4}$ hour. Assume he walked the entire time at the same rate. What was his speed in miles per hour? Priya answered this question, but made a mistake. Explain Priya's mistake. Then complete the work correctly to answer the question.

I need to find: $\frac{2\frac{1}{4} \text{ miles}}{\frac{3}{4} \text{ hour}}$

$\frac{2\frac{1}{4} \text{ miles}}{\frac{3}{4} \text{ hour}} = \frac{3}{4} \div \frac{3}{4}$

$= \frac{3}{4} \cdot \frac{4}{3}$

$= \frac{12}{12}$

= 1 mile per hour

Mistake: Priya changed $2\frac{1}{4}$ to $\frac{3}{4}$ by adding 2 + 1. To write a mixed number as an improper fraction, multiply the whole number by the denominator and add the numerator: 2 × 4 + 1 = 9, so $2\frac{1}{4} = \frac{9}{4}$. Correct: $\frac{9}{4} \times \frac{4}{3} = \frac{36}{12}$ = 3 miles per hour.

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Part III – Problem Solving

The garden club has 6 parents and 15 students. They pull weeds at a park to raise money for new seeds, and a sponsor pays for each bag of weeds they fill. Each parent earned $4.00 per bag and filled one bag every 12 minutes. Each student earned $3.00 per bag and filled 3 bags every 45 minutes. What is the ratio of the amount of money the parents raised in one hour to the amount of money the students raised in one hour?

Parents: one bag every 12 min = 5 bags per hour (60 ÷ 12 = 5); 6 parents × 5 bags × $4 = $120 per hour. Students: 3 bags per 45 min = 4 bags per hour (45 min = $\frac{3}{4}$ hr, 3 ÷ $\frac{3}{4}$ = 4); 15 students × 4 bags × $3 = $180 per hour. Ratio 120 : 180 = 2 : 3 (divide both by 60).